

EIGEN • MATRIX POWERS

Four powers, computed the long way

Each state is the previous one multiplied by A again. Three multiplications to reach A^4 , and every one of them depends on the one before it.

4

1	2	→	9	8	→	41	42	→	209	208
4	3	· A	16	17	· A	84	83	· A	416	417
A			A			A			A	

2

3

4

DETERMINANT

-5, 25, -125, 625 $\det(A^k) = (\det A)^k$

TRACE

4, 26, 124, 626

grows, and by a fixed recurrence

LARGEST ENTRY

4, 17, 84, 417

the growth the trace conceals

WHAT THE INVARIANT COLUMNS SHOW

The determinant runs (-5) and the trace runs $5 + (-1)$ — the eigenvalues of A raised to the power, and nothing else. Neither reads the entries. That is why a closed form exists at all: the growth was never a property of the sixteen numbers on display, only of the two hiding behind them.